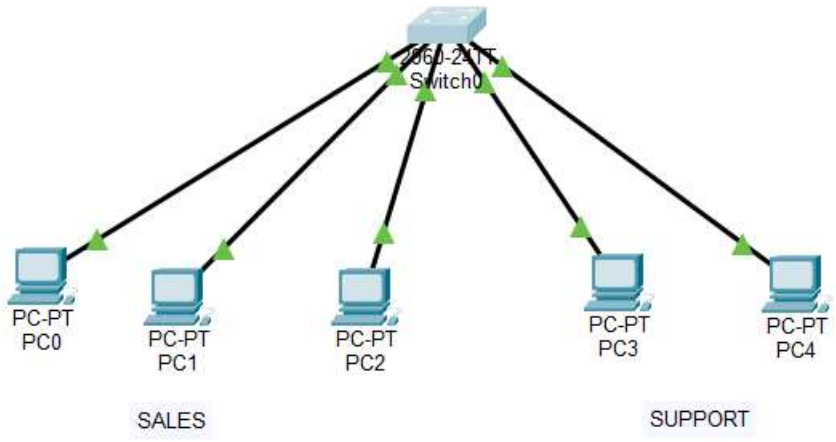
DAY 3

Task 1 Log in to Cisco Packet Tracer or any network simulation tool, add a switch, and create two VLANs named Sales and Support with VLAN IDs 10 and 20.

Diagram



Cisco Switch Commands:

```
enable
configure terminal
vlan 10
name Sales
exit
vlan 20
name Support
exit
end
show vlan brief
```

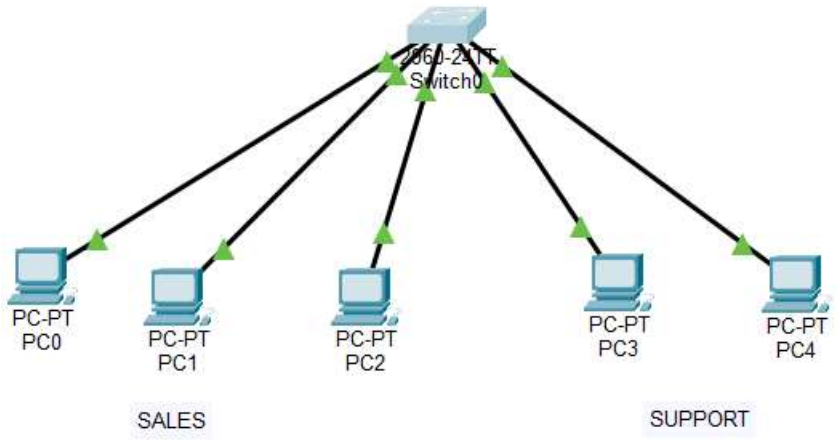
DAY 3

Task 2: Assign three different switch ports to VLAN 10 (Sales) and two ports to VLAN 20 (Support), then verify the VLAN membership using the show vlan brief command.

Commands

```
enable
configure terminal
interface range fastEthernet 0/1 - 3
switchport mode access
switchport access vlan 10
exit
interface range fastEthernet 0/4 - 5
switchport mode access
switchport access vlan 20
exit
end
show vlan brief
```

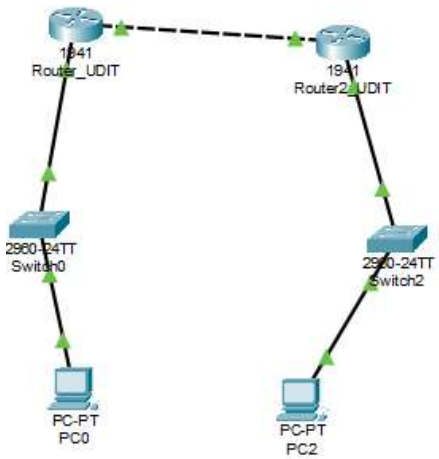
Verification: show vlan brief should list ports Fa0/1, Fa0/2, and Fa0/3 under Sales (VLAN 10), and Fa0/4 and Fa0/5 under Support (VLAN 20). Port numbers may differ depending on the switch model.



DAY 3

Task 3: Connect two routers using a serial or Ethernet link in your simulation tool, assign IP addresses to their interfaces, and configure a static route on Router 1 to reach the LAN behind Router 2.

Network Diagram



Example IP Address Plan

Device/Interface	IP Address	Subnet Mask
Router 1 LAN interface	192.168.1.1	255.255.255.0
Router 1 link to Router 2	10.0.0.1	255.255.255.252
Router 2 link to Router 1	10.0.0.2	255.255.255.252
Router 2 LAN interface	192.168.2.1	255.255.255.0
PC on LAN 1	192.168.1.10	255.255.255.0
PC on LAN 2	192.168.2.10	255.255.255.0

DAY 3

Static Route Commands (Cisco IOS example)

On Router 1:

```
enable
configure terminal
ip route 192.168.2.0 255.255.255.0 10.0.0.2
end
show ip route
```

On Router 2:

```
enable
configure terminal
ip route 192.168.1.0 255.255.255.0 10.0.0.1
end
show ip route
```

Important: Configure the router interfaces with the IP addresses above and enable them with no shutdown. The interface names and link type depend on the router model. Set each PC's default gateway to its local router address.

DAY 3

Task 4: Simulate a scenario where a device in the Sales VLAN tries to ping a device in the Support VLAN and the ping fails. Explain why the ping fails and suggest what configuration would allow inter-VLAN communication.

Question

A device in the Sales VLAN tries to ping a device in the Support VLAN, but the ping fails. Explain why and suggest a solution.

Answer

VLANs create separate Layer 2 broadcast domains. Devices in different VLANs cannot communicate directly through an ordinary Layer 2 switch. Inter-VLAN routing must be configured using a router-on-a-stick setup or a Layer 3 switch.

Suggested Router-on-a-Stick Configuration

On the switch port connected to the router, configure a trunk.

```
interface gigabitEthernet 0/1
switchport mode trunk
end
```

On a router interface connected to the switch (example interface GigabitEthernet 0/0):

```
interface gigabitEthernet 0/0
no shutdown
interface gigabitEthernet 0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
interface gigabitEthernet 0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
```

Then assign PCs in VLAN 10 addresses from 192.168.10.0/24 with gateway 192.168.10.1, and PCs in VLAN 20 addresses from 192.168.20.0/24 with gateway 192.168.20.1. Make sure the VLANs exist and the switch-to-router link is a trunk.

Testing

- Use show vlan brief to confirm VLAN and port assignments.
- Use show interfaces trunk to verify the trunk link.
- Use show ip interface brief to check router interfaces are up.
- Ping the local gateway from each PC.
- Ping from a Sales PC to a Support PC after inter-VLAN routing is configured.

Result

Two VLANs (Sales and Support) were created, switch ports were assigned to each VLAN, and static routes were configured between two routers. Inter-VLAN communication requires an additional Layer 3 routing configuration.